Exp 7 : CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNALS USING MATLAB SIMULATION

Objectives :

1. To generate a Frequency Modulated (FM) signal in MATLAB using specified carrier and message signal parameters.

2. To calculate and verify the frequency deviation of the FM signal from the given carrier frequency and modulating signal parameters.

3. To determine and verify the modulation index of the FM signal using the relationship between frequency deviation and message frequency.

4. To analyze the effect of message frequency on frequency deviation and modulation index, and verify their relationship through MATLAB simulation

Code :

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal

clc;
clear;
close all;

%% Message Signal Parameters
Am = 2; % Message amplitude (V)
fm = 500; % Message frequency (Hz)

%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)

%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.01; % Simulation duration (s)

%% Time Vector
t = 0:1/fs:T;

%% Message Signal
m = Am*cos(2*pi*fm*t);

%% Frequency Deviation
```

```
delta_f = kf*Am;

%% Modulation Index
beta = delta_f/fm;

%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
```

```
Carrier Frequency (fc) = 10000 Hz
```

```
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
```

```
Message Frequency (fm) = 500 Hz
```

```
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
```

```
Message Amplitude (Am) = 2 V
```

```
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
```

```
Frequency Sensitivity (kf) = 250 Hz/V
```

```

fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
```

```
Frequency Deviation (Delta f) = 500.00 Hz
```

```
fprintf('Modulation Index (Beta) = %.2f\n',beta);
```

```
Modulation Index (Beta) = 1.00
```

```

%% Verification
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n',beta_verify);
```

```
Verified Modulation Index = 1.00
```

```

%% Verification Check
tolerance = 1e-6;

if abs(beta - beta_verify) < tolerance
    fprintf('\nVerification Successful\n');
else
    fprintf('\nVerification Failed\n');
end
```

```
Verification Successful
```

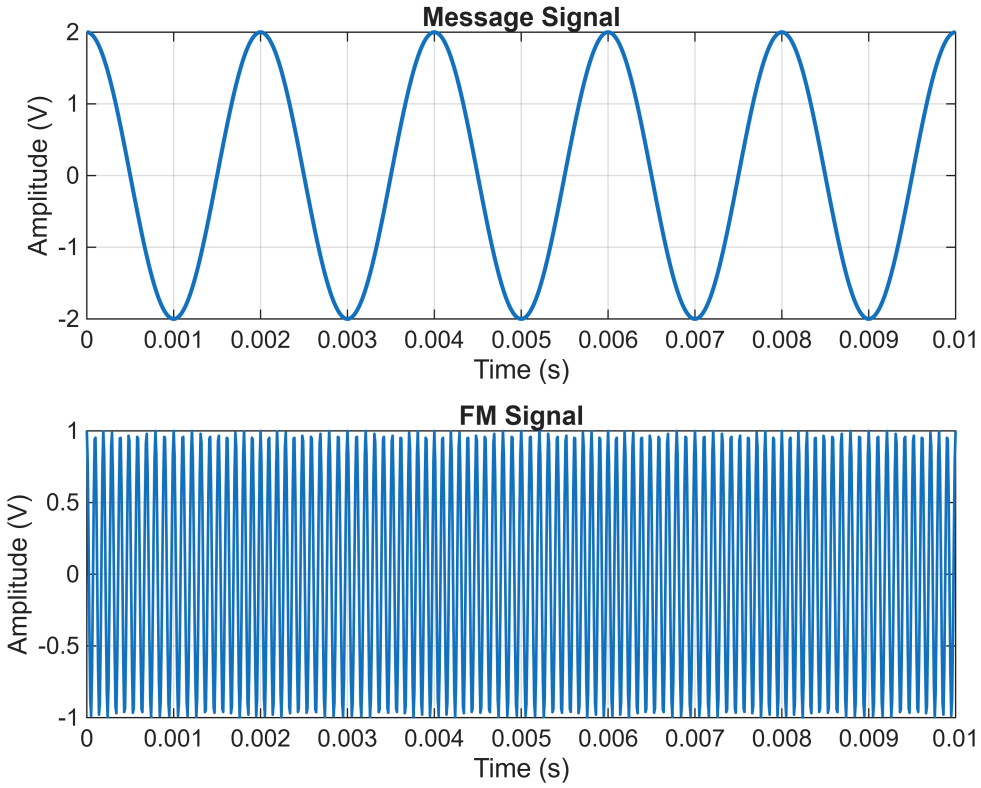
```

%% Plotting
```

```
figure;

subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
```



Result :

1. Generate and visualize carrier, message, and FM signals using MATLAB.

2. Calculate the theoretical frequency deviation and verify it from the simulated FM waveform.

3. Calculate the FM modulation index and verify the relationship β = Δf/fm.

4. Interpret the effect of changing message frequency on modulation index and verify that, for constant

frequency sensitivity and message amplitude, the peak frequency deviation remains constant.